VACATION MINI PROJECT REPORT

STUDY PLANNER

An Interactive Web-Based Academic Task Management System

Project Report submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Computer Applications (BCA)

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Semester	4th Semester
Academic Year	2025 – 2026
Submission	July 2026

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CERTIFICATE

This is to certify that the mini project entitled

STUDY PLANNER

has been successfully developed and submitted by **KIRAN G** (Roll No. **24BCA085**), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru, in partial fulfillment of the academic requirements for the degree of Bachelor of Computer Applications during the Academic Year 2025–2026.

This mini project was developed independently as part of self-learning and skill enhancement during the vacation period. The work presented in this report is original and has not been submitted elsewhere for any academic examination or award.

The project demonstrates a satisfactory level of understanding of web technologies and the ability to design and implement a functional, real-world application.

St. Claret College (Autonomous), Bengaluru

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DECLARATION

I, **KIRAN G**, student of 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru (Roll No. 24BCA085), hereby declare that the mini project report entitled **"STUDY PLANNER"** submitted to the Department of BCA in partial fulfillment of the requirements for the degree of Bachelor of Computer Applications is a record of original work done by me during the vacation period.

I further declare that:

- This project has been carried out by me independently, without any formal institutional guidance, as a self-initiated learning exercise.
- The work presented in this report is genuine and has not been copied or reproduced from any other source without proper acknowledgement.
- This project has not been submitted previously for the award of any degree, diploma, or any other academic title at this or any other institution.
- All references and sources of information used in this project have been duly cited and acknowledged in the appropriate sections.
- The data presented, the analysis performed, and the conclusions drawn are based entirely on my own understanding and research during the project period.

Place: Bengaluru

Date: July 2026

Signature of the Student

KIRAN G

Roll No: 24BCA085

4th Semester, BCA

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and encouraged me throughout the development of this mini project.

I am grateful to the creators and contributors of the open educational resources available through platforms such as MDN Web Docs, W3Schools, Stack Overflow, and YouTube, which served as invaluable references during development. These resources helped me overcome technical challenges and deepen my understanding of HTML, CSS, and JavaScript.

I extend heartfelt thanks to my family for their constant encouragement and for providing a conducive environment for learning during the vacation. Their unwavering support kept me motivated throughout the project.

This project was undertaken entirely on my own initiative as a self-learning exercise, and it has been a truly enriching and rewarding experience. The process of independently planning, designing, developing, and deploying a full-featured web application has significantly enhanced my confidence and technical competence.

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ABSTRACT

In today's fast-paced academic environment, students face increasing difficulty in managing multiple subjects, assignment deadlines, and study schedules simultaneously. The absence of a dedicated, integrated tool for academic task management often leads to missed deadlines, inefficient study habits, and poor academic performance. Conventional approaches such as handwritten planners or generic to-do applications fail to address the specific needs of students.

The **Study Planner** is a web-based academic productivity application designed to address these challenges. It offers students a comprehensive, intuitive platform to manage tasks, maintain study streaks, track progress, and sustain focus through a built-in Pomodoro timer. The application consolidates multiple productivity features into a single, accessible interface that operates entirely on the client side, requiring no installation or login.

Key Features:

- Task management with priority levels (High, Medium, Low) and due-date tracking.
- Subtask support for breaking complex assignments into manageable steps.
- Recurring task configuration (Daily, Weekly) for consistent study habits.
- Pomodoro Timer with configurable work, short-break, and long-break intervals.
- Analytics Dashboard with completion rate charts, last-7-days activity, priority breakdown, and cumulative trend graphs.
- Badges and achievement system to gamify the study experience.
- Daily streak tracking and motivational quote display.

Technologies Used:

HTML5, CSS3, and Vanilla JavaScript. Deployed on Netlify.

Expected Outcome:

The Study Planner is expected to help students build consistent study routines, improve time management, reduce academic stress, and develop a habit of focused, goal-oriented study sessions. The gamification elements such as streaks and badges are designed to sustain long-term engagement with the application.

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CHAPTER 1

INTRODUCTION

1.1 Background

Academic life demands a high degree of self-discipline and time management. Students pursuing degree programmes such as BCA are expected to juggle multiple courses, practical assignments, examinations, and personal study simultaneously. Without a structured system to organise these responsibilities, students often find themselves overwhelmed, leading to procrastination, missed deadlines, and diminished academic performance.

Digital tools have increasingly become a part of academic life, yet most general-purpose productivity applications are not specifically designed with a student's academic workflow in mind. Applications such as Todoist or Notion, while powerful, carry a steep learning curve and are often feature-heavy for simple academic task tracking.

1.2 About the Project

The **Study Planner** is a lightweight, student-centric web application that brings together essential productivity tools in a single, clean interface. Developed using pure front-end technologies — HTML5, CSS3, and JavaScript — it requires no server-side infrastructure, no registration, and no installation. Students can access the application directly through any modern web browser.

The application is available live at: **https://study-planner-gds.netlify.app/** and has been deployed using Netlify's free hosting platform, making it accessible from any device with an internet connection.

1.3 Scope of the Project

This project is designed primarily for undergraduate students and covers the following scope:

- Creation, management, and organisation of academic tasks.
- Visual progress tracking and study analytics.
- Focused study support through the Pomodoro technique.
- Achievement-based motivation through a badge and streak system.
- Fully client-side implementation with browser-based data persistence.

CHAPTER 2

OBJECTIVES OF THE PROJECT

The primary objectives guiding the design and development of the Study Planner are as follows:

1. To provide an integrated task management system: Design a platform where students can create, organise, edit, and track academic tasks in one place, eliminating the need for multiple disjointed tools.

2. To support priority-based task organisation: Enable students to classify tasks by priority level (High, Medium, Low) so that critical assignments receive attention first.

3. To incorporate time-management techniques: Implement the Pomodoro Technique within the application to help students maintain focus and prevent burnout during study sessions.

4. To visualise academic progress through data analytics: Provide a dashboard with graphical representations of task completion rates, priority distributions, daily activity, and cumulative progress trends.

5. To motivate consistent study habits through gamification: Introduce daily streak tracking, achievement badges, and motivational quotes to encourage students to engage with the platform regularly.

6. To develop a fully accessible, cross-platform solution: Build the application using standard web technologies so that it functions seamlessly on desktops, laptops, and mobile devices without any installation.

7. To demonstrate practical application of front-end development skills: Apply core web development concepts including DOM manipulation, local data storage, event-driven programming, and responsive UI design in a real-world context.

CHAPTER 3

EXISTING SYSTEM

3.1 Current Methods Used by Students

Prior to the development of dedicated academic planning tools, students have relied on a combination of traditional and generic digital methods to manage their study schedules. These existing approaches, while widely used, carry significant limitations:

a) Paper-Based Planners and Notebooks

Many students continue to use physical planners, sticky notes, or notebooks to jot down their tasks and deadlines. While this method is straightforward, it offers no automated reminders, no progress tracking, and no visual analytics. Pages can be lost, and there is no way to organise tasks by priority or category efficiently.

b) Generic To-Do Applications

Applications such as Google Keep, Microsoft To-Do, or mobile reminder apps provide basic task listing functionality. However, they lack features specifically tailored to student needs — such as Pomodoro timers, study streaks, academic progress dashboards, or subtask hierarchies.

c) Calendar Applications

Google Calendar or similar tools are used by some students to schedule study sessions. These are time-slot based and do not support task prioritisation, completion tracking, or integrated focus timers.

3.2 Limitations of the Existing System

- Lack of integration: Students must switch between multiple apps for tasks, timers, and calendars.
- No academic-specific features: Generic tools do not support recurring academic tasks, subtasks, or priority-based study queues.
- Absence of analytics: Students have no way to visualise their productivity patterns or completion trends over time.
- No motivation mechanisms: Existing tools do not reward consistent study behaviour or track streaks.
- Data fragmentation: Information is scattered across notebooks, apps, and calendar entries, making holistic planning difficult.
- Platform dependency: Some tools require sign-up or are platform-specific, restricting universal access.

CHAPTER 4

PROPOSED SYSTEM

4.1 Overview

The Study Planner proposes a unified, student-focused web application that consolidates task management, time management, progress analytics, and motivational features into a single, cohesive platform. The system is designed to eliminate the fragmentation associated with existing tools and provide a seamless, distraction-free academic productivity experience.

4.2 Advantages Over the Existing System

- **Integrated Platform:** All study-related tools — task manager, Pomodoro timer, progress charts, and badge system — are available within a single application, removing the need to context-switch between multiple tools.
- **Student-Specific Design:** The interface and feature set are designed specifically around academic workflows, including priority classification, subtask creation, due-date management, and recurring task support.
- **Built-in Analytics:** The Dashboard module provides real-time graphical insights into study patterns, helping students identify their most productive days and track progress towards academic goals.
- **Gamification for Engagement:** Daily streaks, Pomodoro session counts, and achievement badges create a reward loop that encourages consistent usage and sustained study habits.
- **Zero Setup Required:** The application is browser-based and requires no installation, login, or account creation, making it immediately accessible to all students.
- **Offline Capability:** Using browser-based data persistence, the application retains task data between sessions without requiring a network connection.

4.3 System Architecture

The Study Planner follows a client-side-only architecture. All logic, data processing, and storage are handled within the user's browser using JavaScript. The application comprises four primary functional modules — Tasks, Pomodoro, Dashboard, and Badges — each rendering dynamically within a single-page application structure. Data is persisted using the browser's localStorage API, ensuring continuity across sessions.

CHAPTER 5

MODULES AND FEATURES

The Study Planner is structured as a single-page application (SPA) with four distinct functional modules, each accessible via the navigation bar. The following subsections describe each module in detail.

5.1 Global Header and Statistics Bar

The top section of the application remains persistent across all modules and serves as the application's control centre. It displays the current date and a rotating motivational quote sourced from a curated collection. Below the header, a statistics bar presents five key performance indicators:

- **Total:** Total number of tasks created.
- **Completed:** Tasks marked as done.
- **Pending:** Tasks yet to be completed.
- **Streak:** Number of consecutive days on which tasks were completed.
- **Progress:** Overall completion percentage.

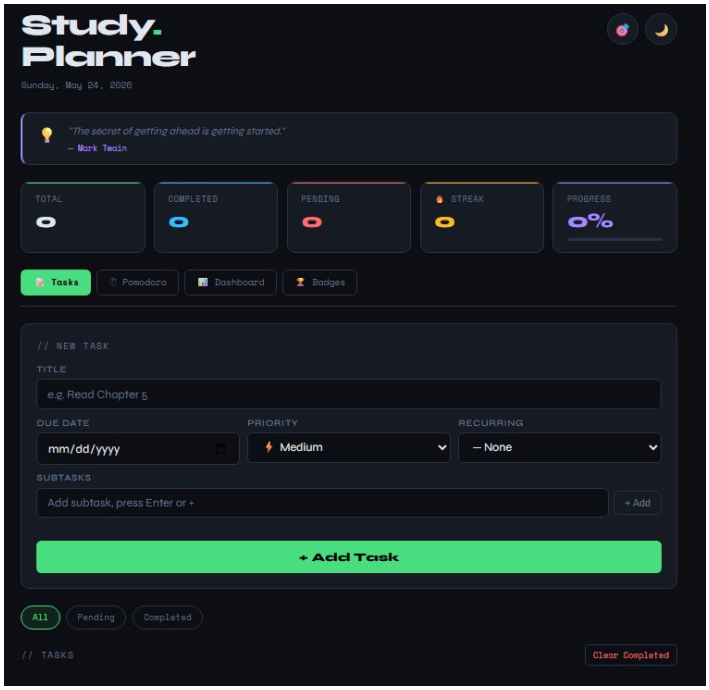


Figure 5.1 – Tasks Module with Statistics Bar and New Task Form

5.2 Tasks Module

The Tasks module is the core of the Study Planner. It provides a full-featured interface for creating, managing, and completing academic tasks.

New Task Form:

The task creation form allows students to enter a task title, set a due date using a date picker, select a priority level (High, Medium, or Low), and configure recurrence (Daily, Weekly, or None). Subtasks can be added within the same form by typing and pressing Enter, enabling hierarchical task breakdown.

Task List View:

Created tasks are displayed in a filterable list supporting three views: All, Pending, and Completed. Each task card shows the title, due date, priority badge, subtask progress, and controls for marking completion or deletion. A 'Clear Completed' button allows batch removal of finished tasks.

5.3 Pomodoro Timer Module

The Pomodoro Timer module implements the well-known time management technique developed by Francesco Cirillo. The technique involves structured work intervals separated by short breaks, promoting sustained concentration.

Timer Configuration:

Students can configure the duration of work sessions, short breaks, and long breaks. They can also set the frequency at which a long break occurs (e.g., after every four Pomodoro cycles). A task can be linked to the timer session for focused tracking.

Focus Mode:

An immersive Focus Mode overlay can be activated, which displays only the timer and the selected task, minimising distractions. The number of Pomodoro sessions completed for the day is tracked and displayed in real time.

5.4 Dashboard Module

The Dashboard provides a visual analytics panel that helps students understand their study patterns and productivity trends.

- **Weekly Insights:** Displays total tasks completed this week, the most productive day, Pomodoros completed today, and high-priority tasks done.
- **Completion Rate Chart:** A doughnut chart comparing completed and pending tasks.
- **Last 7 Days Bar Chart:** Daily task completion visualised as a bar graph for the current week.
- **Priority Breakdown Chart:** A doughnut chart segmenting completed tasks by priority level.
- **Cumulative Trend Line:** A line graph showing cumulative task completion growth over the past week.

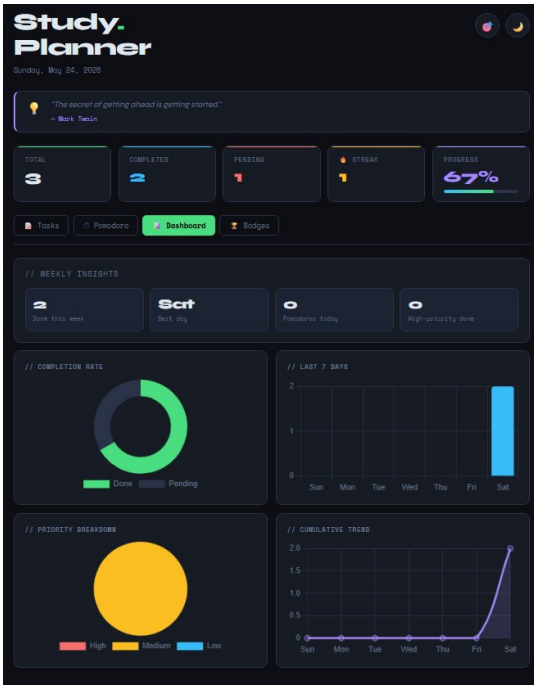


Figure 5.2 – Dashboard Module with Analytics Charts and Weekly Insights

5.5 Badges Module

The Badges module introduces a gamification layer to the application, rewarding students for reaching specific milestones. Each badge has a unique title, icon, and achievement condition.

Badge	Condition
First Step	Add your very first task
3-Day Streak	Complete tasks 3 days in a row
7-Day Streak	Complete tasks 7 days in a row
Getting Going	Complete 10 tasks total
Century	Complete 100 tasks total
Priority Master	Complete 50 high-priority tasks
Pomodoro Fan	Complete 5 Pomodoro sessions today
Focus Machine	Complete 10 Pomodoro sessions today

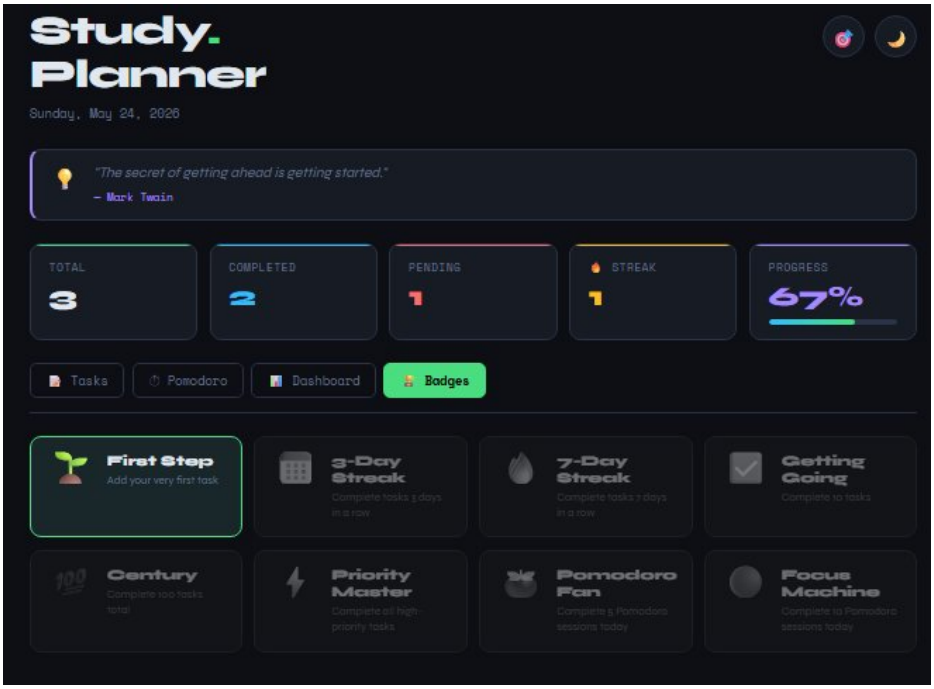


Figure 5.3 – Badges Module Showing Achievement System

CHAPTER 6

TECHNOLOGIES USED

The Study Planner was developed exclusively using front-end web technologies. No back-end server, database, or external API was required. The following technologies were employed:

6.1 HTML5 (HyperText Markup Language 5)

HTML5 forms the structural backbone of the application. Semantic HTML5 elements such as , , , and were used to organise the document structure meaningfully. HTML5 features including the element and custom data attributes were leveraged extensively within the task creation forms.

6.2 CSS3 (Cascading Style Sheets 3)

CSS3 was used to style the entire application, delivering a sleek dark-themed user interface inspired by modern developer tools and productivity applications. Key CSS3 features employed include CSS Custom Properties (variables) for consistent theming, Flexbox and Grid layouts for responsive design, CSS transitions and keyframe animations for interactive feedback, and media queries for mobile responsiveness.

6.3 JavaScript (ES6+)

Vanilla JavaScript served as the application's primary programming language. The project demonstrates extensive use of modern JavaScript features including arrow functions, template literals, destructuring, spread operators, and async event handling. Core application logic includes DOM manipulation for dynamic rendering, the setTimeout/setInterval APIs for timer functionality, and the Date object for streak and deadline calculations.

6.4 localStorage API

The Web Storage API's localStorage interface was used to persist task data, badge states, Pomodoro session counts, and streak information across browser sessions. This eliminates the need for any server-side storage while maintaining data continuity for the user.

6.5 Canvas API / Chart Rendering

The Dashboard module's charts (doughnut charts, bar charts, and line graphs) were rendered using JavaScript's Canvas API or a lightweight charting library, enabling dynamic, data-driven visualisations within the browser.

6.6 Netlify (Deployment Platform)

The application was deployed and hosted on Netlify, a cloud-based static site hosting platform. Netlify provides continuous deployment, a free custom subdomain, and global CDN distribution, making the application fast and accessible worldwide at no cost.

#	Technology	Version / Standard	Purpose
1	HTML	HTML5	Page structure and semantic markup
2	CSS	CSS3	Styling, layout, theming, animations
3	JavaScript	ES6+	Application logic, DOM, timers, events
4	localStorage	Web Storage API	Client-side data persistence
5	Canvas/Charts	HTML5 Canvas	Dashboard visualisations
6	Netlify	Cloud Platform	Deployment and hosting

CHAPTER 7

CONCLUSION

The Study Planner project represents a meaningful attempt to address a genuine and widely experienced challenge in student life — the difficulty of organising academic tasks, sustaining focused study sessions, and maintaining consistent productivity over an academic term. By combining task management, the Pomodoro technique, visual analytics, and a gamification-based reward system within a single, accessible web application, the project delivers a practical and student-friendly productivity tool.

From a technical perspective, the project demonstrates the expressive power of core front-end technologies. Without relying on any framework or back-end infrastructure, the application achieves a feature set comparable to commercial productivity tools. The use of HTML5, CSS3, and vanilla JavaScript reinforced foundational programming concepts including event-driven architecture, modular function design, dynamic DOM rendering, and client-side state management through localStorage.

The Dashboard module, in particular, highlights the capability of JavaScript to process data and render meaningful visualisations without external dependencies. The badge and streak systems demonstrate how application design can be leveraged to influence user behaviour in a positive, goal-oriented direction.

On a personal level, undertaking this project independently during the vacation period was an immensely valuable learning experience. Planning the feature set, designing the user interface, writing and debugging the logic, and finally deploying the application on Netlify gave a comprehensive, end-to-end view of the web development lifecycle. It reinforced the importance of iterative development, user-focused design thinking, and the discipline of self-directed learning.

In conclusion, the Study Planner is a well-rounded, functional mini project that successfully achieves its stated objectives. It stands as a testament to the value of self-initiated learning and the practical application of theoretical knowledge gained through the BCA programme. The project is publicly accessible and ready for further development as the developer's technical skills continue to grow.

Live Project: https://study-planner-gds.netlify.app/